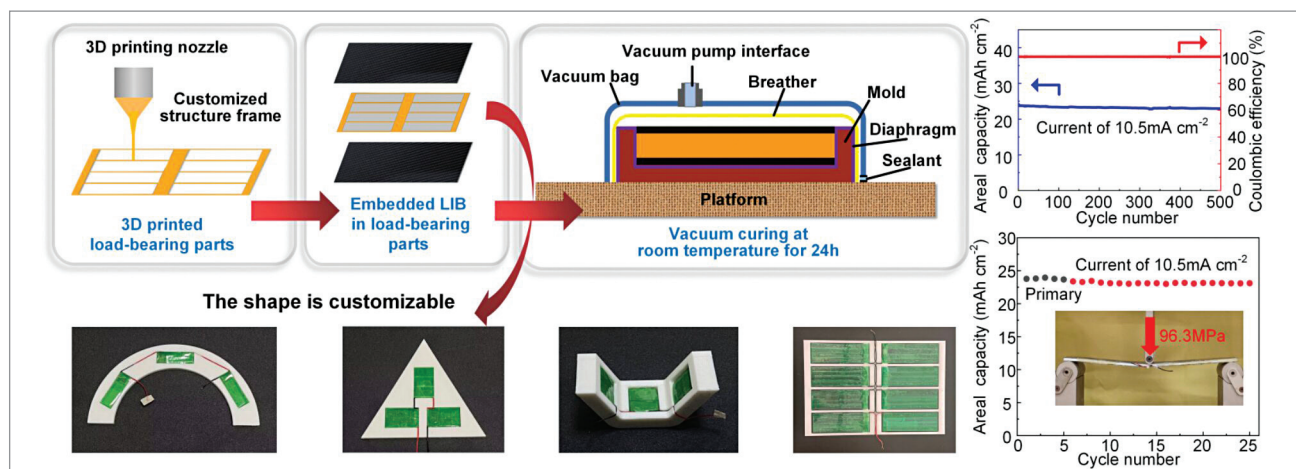


RESEARCH & INNOVATION UPDATES

3D lithium-ion batteries in customisable shapes



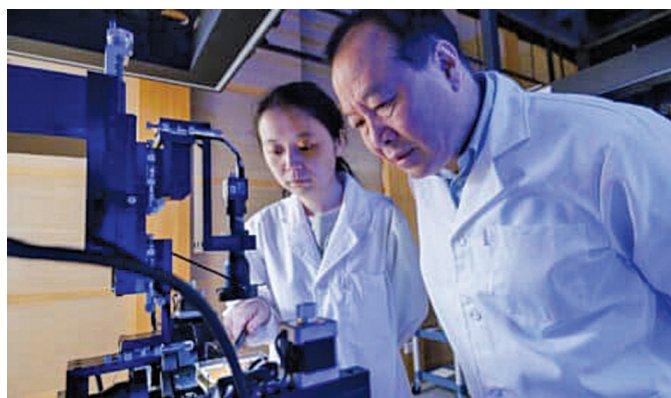
3D structural lithium-ion batteries (Credit: Composites Science and Technology, 2024)

The future of electric vehicles (EVs) hinges on advancements in fast-charging, efficient batteries that support long operations. A promising innovation is the development of structural batteries, which serve as both energy storage systems and integral parts of the vehicle's structure. Unlike traditional external batteries, these are embedded into the vehicle's design, enhancing efficiency.

Researchers at Shanghai University have developed 3D-printed structural lithium-ion batteries with customisable shapes, overcoming previous limitations of low energy density and poor performance. This technology, which integrates energy storage with structural support, has potential applications in EVs, autonomous robots, and warehouse logistics vehicles.

Quantum chip simplifies molecule study

Researchers at The Hong Kong Polytechnic University (PolyU), in collaboration with members from Nanyang Technological University, City University of Hong Kong, Beijing Institute of Technology, Southern University of Science and Technology, the Institute of Microelectronics, and Chalmers University of Technology in Sweden, have introduced a 16-qubit quantum microprocessor chip. This chip, designed specifically for the simulation of molecular spectroscopy for large and complex molecules, is part of a comprehensive system that includes optical-electrical-thermal packaging and a fully programmable control module and software. This development marks a significant step forward in quantum chemistry and computational applications, exceeding the capabilities of classical computing. Such quantum technology paves the way for new applications in molecular docking and quantum machine learning techniques like graph classification.



Researchers at PolyU have introduced a quantum microprocessor chip for simulating molecular spectroscopy in complex molecules, a global first (Credit: <https://www.polyu.edu.hk>)